

Practice Questions for 12L2-distance-clusters

Generated by Scribe.AI

December 18, 2024

1 Questions

1. According to the lecture, what is the primary goal of clustering?
 - A. To predict future outcomes based on labeled data
 - B. To group similar objects into classes
 - C. To reduce the dimensionality of the data
 - D. To find the optimal parameters for a model
 - E. To classify data into predefined categories
2. According to the lecture, what is the most commonly used distance measure for continuous data in clustering?
 - A. Hamming distance
 - B. Matching coefficient
 - C. Euclidean distance
 - D. Manhattan distance
 - E. Cosine similarity
3. According to the lecture, what type of clustering is K-means?
 - A. Hierarchical, hard clustering
 - B. Hierarchical, soft clustering
 - C. Flat, hard clustering
 - D. Flat, soft clustering
 - E. None of the above

2 Answers

1. According to the lecture, what is the primary goal of clustering?
 - A. Answer A is incorrect because predicting future outcomes is a goal of supervised learning, not clustering.
 - B. Answer B is correct because clustering is defined as the process of grouping a set of objects into classes of similar objects.
 - C. Answer C is incorrect because dimensionality reduction is a separate task, although it can be used in conjunction with clustering.
 - D. Answer D is incorrect because finding optimal parameters is a goal of model training, not clustering.
 - E. Answer E is incorrect because classifying data into predefined categories is a goal of supervised learning, not clustering.
2. According to the lecture, what is the most commonly used distance measure for continuous data in clustering?
 - A. Answer A is incorrect because Hamming distance is used for discrete data, not continuous data.
 - B. Answer B is incorrect because the matching coefficient is used for discrete data, not continuous data.
 - C. Answer C is correct because the lecture states that Euclidean distance is the most commonly used distance measure for continuous data.
 - D. Answer D is incorrect because while Manhattan distance is a valid distance measure, Euclidean distance is more commonly used for continuous data.
 - E. Answer E is incorrect because cosine similarity is typically used for text or high-dimensional data, not general continuous data.
3. According to the lecture, what type of clustering is K-means?
 - A. Answer A is incorrect because K-means is not hierarchical.
 - B. Answer B is incorrect because K-means is not hierarchical and is not soft clustering.
 - C. Answer C is correct because the lecture states that K-means is a flat, hard clustering approach.
 - D. Answer D is incorrect because K-means is not soft clustering.
 - E. Answer E is incorrect because K-means is a flat, hard clustering approach.